

Recap

The Page Load Time is the metric used to assess the efficiency of web applications

It includes the time for the TCP and TLS handshakes, the time for the HTTP request and response and the time to transfer the requested object

To reduce the TCP/TLS overheads for downloading multiple objects stored on a given web server, TCP connections can be reused, i.e., persistent connections

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Non persistent vs persistent connections

Non-persistent connections

TCP connection closed after each response message

Each object requires a new TCP connection, that is, two RTTs plus the operating system overhead (data structures to track connection state, timings and transitions)

Persistent connections

TCP connection not closed after receiving a response

Subsequent HTTP requests/responses between a client and a server are sent over the open TCP connection

A client sends a request as soon as it encounters a referenced object (one RTT per object)

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Example

Analysis of the number of RTTs requested to download web pages using

- a) non persistent TCP connections
- b) persistent TCP connections

Page with

- 1. no referenced objects
- 2. one referenced object
- 3. two referenced objects
- 4. three referenced objects
- 5. ten referenced objects
- 6. 50 referenced objects
- 7. 100 referenced objects
- 8. 1000 referenced objects

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Sequential vs parallel transfers

Sequential transfers affect efficiency and PLT

Browsers can open multiple concurrent TCP connections (HTTP/1.1)

Maximum number of persistent connections per server specified in the browser settings [user preference]

Browser settings must match web server settings (e.g., max number of open connections per client/IP set by servers to avoid overload and monopolization of few clients)

This solution improves PLT at the expense of increased resource usage

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Google Drive home page: distributed content

```
<head>
<meta charset="utf-8" />
<title>Google Drive: Share Files Online with Secure Cloud Storage | Google Workspace</title>
<meta name="description" content="Learn about Google Drive's file sharing platform that provides personal, secure cloud storage option to share content with other users." />
<link rel="canonical" href="https://workspace.google.com/products/drive/" />
<link href="https://www.gstatic.com/images/branding/googlegradient/1x/googlegradient_standard_20dp.png" rel="shortcut icon" type="image/png" />

<link rel="stylesheet" href="https://fonts.googleapis.com/css?family=Material+Icons+Outlined&display=swap" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/SdHIXrh5.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/zPypI3Wr.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/BEwVz9nn.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/8nwFsvE3.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/DYv9Fyjj.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/CZx98z6f.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/Bctcio4i.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/DB6tyjNO.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/7wYUZQKH.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />
<link rel="stylesheet" href="/assets/Colj0pNL.css" nonce="1J-AH-9gMuqt01JMDBrC8A" />

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```

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HTTP messages

HTTP messages are the basic unit of communication between clients and servers

A message is a sequence of bytes with a *type*, *syntax* and *semantic*

Two types of messages:

- HTTP request messages (sent by clients)
- HTTP response messages (sent by web servers)

Each message includes:

- Control data that describes the primary purpose of the message
- Header fields, i.e., name/value pairs for extending control data and conveying additional information about sender, message, content or context
- [Optional] content

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HTTP messages (cont'd)

Control data of request messages includes a request method, a request target (i.e., URI) and an (optional) protocol version

Control data of response messages includes a status code, an (optional) reason phrase and the protocol version used

Syntax slightly varies across the HTTP versions

In HTTP/1.1 control data is the first line of the request/response messages

In HTTP/2 and HTTP/3, control data is included in the header section of request/response messages, using pseudo-header fields with a reserved name (e.g., **:scheme**, **:status**)

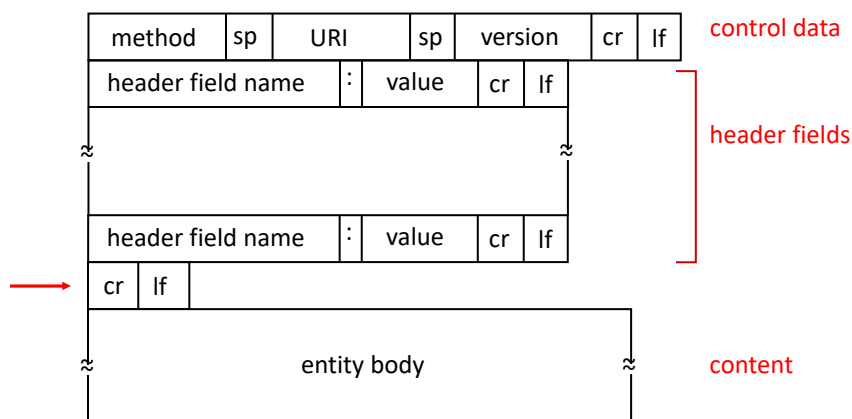
Each line is delimited by two characters: *carriage return* (ASCII code=13) + *line feed* (ASCII code=10), i.e., CRLF

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Format of HTTP/1.1 request messages



For HTTP/2 and HTTP/3 control data is associated with pseudo-headers

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HTTP/1.1 request messages: Firefox & Chrome

GET / HTTP/1.1

Host: webing.unipv.eu
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:149.0) Gecko/20100101 Firefox/149.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Accept-Encoding: gzip, deflate
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Priority: u=0, i
Pragma: no-cache
Cache-Control: no-cache

GET / HTTP/1.1

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9
Cache-Control: no-cache
Connection: keep-alive
Host: webing.unipv.eu
Pragma: no-cache
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36

No content

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HTTP/2 request message: Microsoft Edge

:authority portale.unipv.it
:method GET
:path /en
:scheme https

Accept text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding gzip, deflate, br, zstd
Accept-Language en-US,en;q=0.9,it;q=0.8,en-GB;q=0.7
Cache-Control no-cache
Cookie __cf_bm=JdlPiwl9GDupE7ob0j7E_tYHNLXsgvypBabGSaIQfCM-1776258088.8971455-1.0.1.1-2bv88w3fQXCJr4IVARm6wil.qCYjFVhkDO0NP0NC7utgP2N1hs.GXLnjU2VvB_xrNFWvn3CZ0_Q12G3QOsDd3g2EwUgiHFnexHtYEv512mZB2gBNLTV92MFnwLuMtnMD;nmstat=76c21686-39db-a30c-5366-b03854909f08
Pragma no-cache
Priority u=0, i
Sec-Ch-Ua "Microsoft Edge";v="147", "Not.A/Brand";v="8", "Chromium";v="147"
Sec-Ch-Ua-Mobile ?0
Sec-Ch-Ua-Platform "Windows"
Sec-Fetch-Dest document
Sec-Fetch-Mode navigate
Sec-Fetch-Site none
Sec-Fetch-User ?1
Upgrade-Insecure-Requests 1
User-Agent Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0

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HTTP request methods

A method defines the action to be performed on the object specified by control data

Actions can be subdivided in four categories:

- Download:
 - **GET** (entire object)
 - **HEAD** (headers associated with object)
- Upload:
 - **POST** (data to a process)
 - **PUT** (entire object)
 - **PATCH** (partial changes)
- Delete:
 - **DELETE** (entire object)
- Miscellanea:
 - **OPTIONS** (info about communication options)
 - **TRACE** (creates a loop back for debugging)
 - **CONNECT** (opens tunnels via proxy servers)

body is empty

body is not empty

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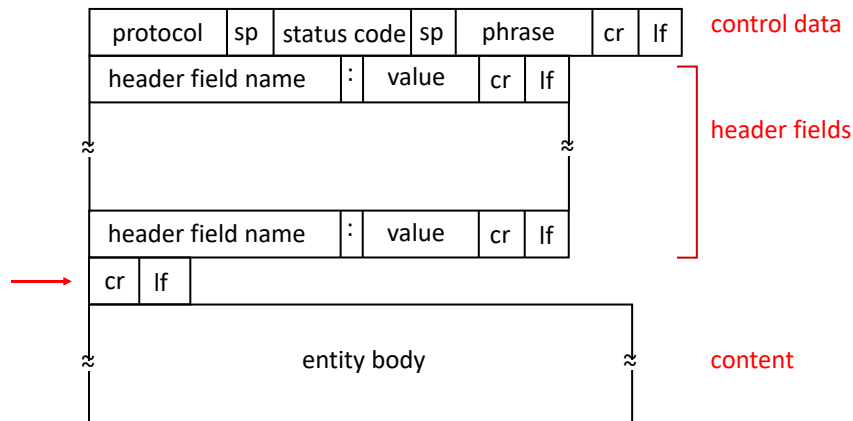
Component	Value
:authority	www.google.com
:method	GET
:path	/search?q=network+sniffing&sca_esv=b23efcb65468375a&source=hp&ei=5l3fabS_HpqI9u8P_6LP-AE8ifIsig=AFdpzrgAAAAAd-b9CdqKl0IZcxhMwqneDzeyf9mm7UU&ved=0ahUKewi0we_W9u-TAxUahP0HHX9RGh8Q4dUDCBQ&uact=5&oq=network+sniffing&gs_lp=Egdnnd3Mtd2l6lhBuZXR3b3JrIHNaWZmaW5nMgUQABiABDIFEAAyGAYyBRAAGIAEMgUQABiABDIFEAAyGAYyBRAAGIAEMgUQABiABDIFEAAyGAYyCxAAGIAEGloFGIYDSOFzUONiWNZwcAR4AJABAjgBlgGgAYEOqgEDOC45uAEDyAEA-AEBmAIUoAKzD6gCAMICBRAuGIAEWglLEC4YgAQYxwEY0QPCAggQABiAB8BiiBMICBxAAGIAEGA3CagYQABgeGA2YAwPxBX3_xmKaXPG2kgcNi4xNKAHymyyBwQzLjE0uAefD8IHbZuAuNi42LjIjB4UBgAgB&scslent=gws-wiz
:scheme	https

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Format of HTTP/1.1 response messages



For HTTP/2 and HTTP/3 control data is associated with a pseudo-header
:status

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HTTP response messages: Mozilla Firefox

```
HTTP/1.1 200 OK
Date: Wed, 15 Apr 2026 12:46:54 GMT
Server: Apache/2.2.16 (Debian)
Link: <http://webing.unipv.eu/wp-json/>; rel="https://api.w.org/"
Vary: Accept-Encoding
Keep-Alive: timeout=15, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=UTF-8
X-Content-Encoding-Over-Network: gzip
Transfer-Encoding: chunked
```

BODY

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Other HTTP response messages

HTTP/1.1 301 Moved Permanently

Date: Wed, 15 Apr 2026 13:25:38 GMT
Server: Apache/2.4.38 (Debian)
Location: https://portale.unipv.it/
Content-Length: 312
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=

No content is associated with these response messages (body is empty)

HTTP/2 301

date: Wed, 15 Apr 2026 13:26:48 GMT
content-type: text/html; charset=utf-8
location: https://portale.unipv.it/it
server: cloudflare
x-drupal-route-normalizer: 1
content-language: it
x-content-type-options: nosniff
x-frame-options: SAMEORIGIN
cache-control: public, max-age=14400
vary: Cookie
expires: Wed, 15 Apr 2026 17:26:48 GMT
x-drupal-cache: HIT
last-modified: Wed, 15 Apr 2026 11:02:57 GMT
x-varnish: 234685445
x-cache: MISS
grace: none
server-timing: cfCacheStatus;desc="HIT"

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